Light

Network Protocols & Internet

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Edexcel GCSE Computer Science — **Network Protocols and Layers (1CP2/01 Topic 4: Networks)**. This topic covers what protocols are, common protocols used on networks and the internet (Ethernet, Wi-Fi, TCP/IP, HTTP, HTTPS, FTP, POP3, SMTP, IMAP), and the four-layer TCP/IP model.

Exam questions typically ask you to **state** what a protocol is, **identify** which protocol is used for a given purpose, **describe** the function of each TCP/IP layer, and **explain** how protocols enable communication between devices. The 2024 Paper 1 included questions on FTP and protocol layers. Expect questions worth 1—4 marks requiring you to match protocols to purposes and describe layer functions.

Why network protocols matter

Protocols are essential for all network communication — without agreed rules, devices could not exchange data:

- **Universal communication:** Protocols allow devices from different manufacturers to communicate
- **Internet foundation:** Every website visit, email, and file download relies on protocols

- **Layered design:** The TCP/IP model organises protocols into layers, each with a specific role
- **Security:** Protocols like HTTPS ensure data is encrypted during transmission
- **Exam relevance:** This topic appears in Paper 1 as part of Topic 4, typically worth 4—8 marks across questions

What you must know

This checklist covers the key exam-visible concepts from Edexcel 1CP2 Topic 4: Networks. You should be able to explain each point and apply it to exam scenarios.

- **Define** what a protocol is and explain why protocols are needed for network communication
- **Identify** which protocol is used for a given purpose: Ethernet, Wi-Fi, TCP/IP, HTTP, HTTPS, FTP, POP3, SMTP, IMAP
- **Describe** the purpose of each of the four layers in the TCP/IP model: application, transport, internet, and link
- **Explain** how protocols at different layers work together to enable communication
- **Compare** HTTP and HTTPS in terms of security
- **Describe** the difference between POP3 and IMAP for retrieving email

> **Specification link:** This topic covers Edexcel 1CP2 Paper 1 Topic 4: Networks — specifically protocols (Ethernet, Wi-Fi, TCP/IP, HTTP, HTTPS, FTP, POP3, SMTP, IMAP) and the 4-layer TCP/IP model.

Toolkit

These are the key patterns, definitions and tables you need to apply in exam questions about protocols.

Protocol definition

Protocol: A set of rules that govern how data is transmitted and received across a network.
Key phrase for exams: "A protocol is a **set of rules** that defines how devices **communicate** on a network."

Protocol purposes at a glance

Protocol	Purpose	Exam trigger phrase
Ethernet	Wired LAN communication	"wired connection", "LAN cable", "physical network"
Wi-Fi	Wireless LAN communication	"wireless connection", "without cables"
TCP/IP	Foundation protocol suite for internet communication	"internet communication", "packet delivery"
HTTP	Transferring web pages (unsecured)	"web page", "website", "browser"

Protocol	Purpose	Exam trigger phrase
HTTPS	Transferring web pages securely (encrypted)	"secure website", "encrypted", "online banking"
FTP	Transferring files between computers	"upload files", "download files", "file transfer"
SMTP	Sending email	"send email", "outgoing mail"
POP3	Receiving email (downloads and deletes from server)	"receive email", "download email", "single device"
IMAP	Receiving email (keeps on server, syncs across devices)	"sync email", "multiple devices", "webmail"

TCP/IP four-layer model

```
┌──────────────────────────────────────────────────────────────────────┐ │ LAYER
4: APPLICATION LAYER │ │ Protocols: HTTP, HTTPS, FTP, SMTP, POP3, IMAP │ │
Function: Provides services to user applications │
├──────────────────────────────────────────────────────────────────────┤ │ LAYER
3: TRANSPORT LAYER │ │ Protocols: TCP │ │ Function: Splits data into
packets, ensures reliable delivery │
├──────────────────────────────────────────────────────────────────────┤ │ LAYER
2: INTERNET LAYER │ │ Protocols: IP │ │ Function: Adds IP addresses, routes
packets across networks │
├──────────────────────────────────────────────────────────────────────┤ │ LAYER
1: LINK LAYER (Network Interface) │ │ Protocols: Ethernet, Wi-Fi │ │
Function: Handles physical transmission on local network │
└──────────────────────────────────────────────────────────────────────┘
```

Layer functions summary

Layer	Function (exam phrase)	Example protocols
Application	Provides network services to applications; handles user requests	HTTP, HTTPS, FTP, SMTP, POP3, IMAP
Transport	Splits data into packets; ensures reliable delivery; manages connections	TCP
Internet	Adds IP addresses; routes packets across networks to destination	IP
Link	Converts packets to signals; handles physical transmission on local network	Ethernet, Wi-Fi

1. What are protocols?

A **protocol** is a set of rules that governs how data is transmitted and received across a network. Without agreed protocols, devices would not be able to understand each other — like two people trying to have a conversation in different languages with no common words.

1.1 Why protocols are needed

Networks connect devices made by different manufacturers, running different operating systems, in different locations. Protocols ensure they can all communicate by defining:

- **Format:** How data is structured and organised
- **Timing:** When data should be sent and how fast
- **Sequencing:** The order in which data is transmitted
- **Error handling:** How to detect and correct transmission errors
- **Addressing:** How to identify the source and destination

Because everyone follows the same rules, a Windows computer can communicate with a Mac, a phone can access a website hosted on a Linux server, and email can travel between different email providers.

> **Exam tip:** When asked why protocols are needed, focus on the idea of a "common set of rules" that allows different devices to communicate — like a shared language.

1.2 Protocol standardisation

Protocols are standardised so that all manufacturers and software developers follow the same rules. This is why:

- You can use any web browser to access any website
- You can send email from Gmail to Outlook
- You can connect any device to a Wi-Fi network

Without standardisation, the internet would not work — each manufacturer would have their own incompatible rules.

1.3 Types of protocols

Different protocols handle different aspects of network communication:

- **Physical/link protocols:** Handle how data is transmitted over cables or wireless signals (Ethernet, Wi-Fi)
- **Network protocols:** Handle addressing and routing data across networks (IP)
- **Transport protocols:** Handle splitting data into packets and ensuring delivery (TCP)
- **Application protocols:** Handle specific services like web browsing, email, file transfer (HTTP, SMTP, FTP)

1.4 How protocols enable communication

Imagine two people trying to communicate across a busy room. They need to agree on several things: what language to speak, how loud to talk, how to signal when they want to speak, and what to do if they cannot hear each other. Network protocols work the same way.

When your computer sends data to a web server:

- The application protocol (HTTP) defines the format of the request
- The transport protocol (TCP) splits the data and ensures it arrives correctly
- The network protocol (IP) addresses and routes the data
- The link protocol (Ethernet/Wi-Fi) transmits the signals physically

Each protocol handles its specific job, and together they enable reliable communication between any devices on the network.

1.5 Real-world analogy: postal system

Protocols work like the rules of a postal system:

Postal system	Network equivalent	Protocol role
Writing a letter in English	Formatting a web request	Application layer (HTTP)
Putting letter in envelope, adding tracking	Splitting into packets, adding sequence numbers	Transport layer (TCP)
Writing address on envelope	Adding IP addresses	Internet layer (IP)
Postman collecting and delivering	Physical transmission on cables/wireless	Link layer (Ethernet/Wi-Fi)

Just as the postal system needs standard address formats, sorting procedures, and delivery methods, computer networks need standardised protocols at each stage of communication.

Worked Example 1 — Defining a protocol

Question: Define what is meant by a network protocol. [2 marks]

Model answer:
A protocol is a set of rules **(1)**.
It defines how data is transmitted/communicated across a network **(1)**.

Exam tip: For 2 marks, include both elements: (1) set of rules, (2) governs data transmission/communication. Simply saying "rules for the internet" is too vague.

Worked Example 2 — Explaining why protocols are needed

Question: Explain why protocols are needed for devices to communicate on a network. [2 marks]

Model answer:
Protocols provide a common set of rules that all devices follow **(1)**.
This allows devices from different manufacturers/running different software to understand each other and communicate successfully **(1)**.

Exam tip: Link the "common rules" to the outcome — devices can communicate even if they are different. Think of it like a shared language.

Worked Example 3 – Protocol standardisation

Question: State one benefit of having standardised protocols. [1 mark]

Model answer:
Devices from different manufacturers can communicate with each other **(1)**.
Also acceptable: Any web browser can access any website / Email can be sent between different providers / Ensures compatibility between devices

Exam tip: "State" requires a brief answer. Focus on interoperability — different systems working together because they follow the same rules.

Worked Example 4 – Types of protocols

Question: A company uses both wired and wireless connections. Name one protocol used for wired LAN communication and one protocol used for wireless LAN communication. [2 marks]

Model answer:
Wired: Ethernet **(1)**.
Wireless: Wi-Fi **(1)**.

Exam tip: Ethernet = wired LAN. Wi-Fi = wireless LAN. Do not confuse these with internet protocols like HTTP or IP.

2. Common network protocols

The Edexcel specification requires you to know the purpose of several specific protocols: Ethernet, Wi-Fi, TCP/IP, HTTP, HTTPS, FTP, POP3, SMTP, and IMAP. This section covers each protocol and when it is used.

2.1 Link layer protocols

These protocols handle communication on the local network — the physical transmission of data:

Ethernet

Ethernet is a protocol for **wired LAN communication**. It defines how data is transmitted over cables (typically using RJ45 connectors) and how devices share the network medium.

- Uses **MAC addresses** to identify devices on the local network

- Defines how data is packaged into frames for transmission
- Handles collision detection if two devices transmit simultaneously

Wi-Fi

Wi-Fi is a protocol for **wireless LAN communication**. It defines how data is transmitted using radio waves through wireless access points.

- Uses **MAC addresses** like Ethernet
- Defines radio frequencies and channels
- Includes security standards (WPA2, WPA3) for encrypted connections

> **Exam tip:** Both Ethernet and Wi-Fi operate at the link layer — they handle local network communication. The difference is wired vs wireless transmission.

2.2 TCP/IP

TCP/IP (Transmission Control Protocol/Internet Protocol) is the foundation protocol suite for the internet. It is actually two protocols working together:

IP (Internet Protocol)

IP handles **addressing and routing**:

- Assigns unique IP addresses to devices (e.g., 192.168.1.1)
- Routes packets across networks to reach the destination
- Does not guarantee delivery — it is "best effort"

TCP (Transmission Control Protocol)

TCP handles **reliable delivery**:

- Splits data into smaller packets for transmission
- Numbers packets so they can be reassembled in order
- Requests retransmission of lost or corrupted packets
- Establishes a connection before sending data

> **Together:** IP addresses and routes packets, while TCP ensures they arrive correctly and in order. This is why the internet is reliable even though it consists of millions of interconnected networks.

2.3 Understanding packets

When data is sent across a network, it is not sent as one continuous stream. Instead, it is broken into small **packets**. This is a fundamental concept for understanding how protocols work.

> #### Why use packets?
> - **Efficiency:** Multiple communications can share the same network links
> - **Reliability:** If one packet is lost, only that packet needs retransmitting (not the whole message)
> - **Routing:** Packets can take different routes to avoid congestion
> - **Error handling:** Errors in one packet do not affect others

Each packet contains:

- **Header:** Information about the packet (source address, destination address, sequence number, protocol type)

- **Payload:** The actual data being transmitted
- **Trailer:** Error checking information (in some protocols)

2.4 Web protocols – HTTP and HTTPS

HTTP (HyperText Transfer Protocol)

Purpose: Transferring web pages between web servers and browsers.

- Used when you type a URL or click a link
- Browser sends HTTP request; server returns HTTP response with the page
- **Not encrypted** — data can be intercepted and read
- Suitable for non-sensitive public content

HTTPS (HTTP Secure)

Purpose: Transferring web pages securely using encryption.

- Uses encryption to protect data during transmission
- Data cannot be read even if intercepted
- Shows padlock icon in browser address bar
- **Essential for:** Online banking, shopping, login pages, any site handling personal data

How HTTP requests work

When you visit a website, your browser and the web server communicate using HTTP. This follows a request-response pattern:

Step	Direction	What happens
1	Browser → Server	Browser sends HTTP request (e.g., "GET /index.html")
2	Server processes	Server finds the requested page/resource
3	Server → Browser	Server sends HTTP response with the page content
4	Browser renders	Browser displays the web page to the user

Why HTTPS matters

Without encryption (HTTP), anyone who can intercept your network traffic can read:

- Passwords you enter on login forms
- Credit card details on shopping sites
- Personal messages and emails
- Any data you submit to websites

HTTPS prevents this by encrypting all data before transmission. Even if intercepted, the data appears as meaningless characters to anyone without the decryption key.

> **Exam tip:** The key difference between HTTP and HTTPS is **encryption**. HTTPS encrypts data so it cannot be read if intercepted. This is essential for sensitive

information like passwords and payment details.

2.4 File transfer – FTP

FTP (File Transfer Protocol)

Purpose: Transferring files between computers over a network.

- Used to upload files to web servers (e.g., website content)
- Used to download files from remote servers
- Typically requires username and password authentication
- Can transfer large files that would be impractical via email

Common uses: Web developers uploading website files, businesses exchanging large documents, downloading software from servers.

FTP in practice

FTP uses a client-server model. The FTP client (software on your computer) connects to an FTP server (software on the remote computer). Once connected, you can:

- Browse directories on the remote server
- Upload files from your computer to the server
- Download files from the server to your computer
- Rename, delete, or move files (if you have permission)

Exam scenario: "A web developer has created a new website on their local computer. They need to make it available on the internet." The answer is FTP — the developer uploads the website files to a web server using FTP.

> **Note:** Standard FTP does not encrypt data. For secure file transfer, SFTP (SSH File Transfer Protocol) or FTPS (FTP Secure) can be used, though these are not required for the Edexcel specification.

2.5 Email protocols – SMTP, POP3, IMAP

Email requires multiple protocols — one to send email and others to receive it:

SMTP

Simple Mail Transfer Protocol

Purpose: Sending outgoing email.

- Used when you click "Send"
- Transfers email from your device to your email server
- Transfers email between mail servers

Remember: SMTP = Sending Mail To People

POP3

Post Office Protocol (v3)

Purpose: Receiving email by downloading.

- Downloads email to your device
- Usually **deletes from server** after download
- Email stored locally on one device
- Good for: Single device access, offline reading

IMAP

Internet Message Access Protocol

Purpose: Receiving email while keeping it on server.

- **Keeps email on server**
- Syncs across multiple devices
- Read on phone = marked read on laptop
- Good for: Multiple devices, webmail

Feature	POP3	IMAP
Email location	Downloaded to device, deleted from server	Kept on server, synced to devices
Multiple devices	Difficult — email only on one device	Easy — same view on all devices
Storage	Uses device storage	Uses server storage
Internet required	No — can read offline after download	Yes — needs connection to access
Best for	Single device, limited server space	Multiple devices, webmail access

Worked Example 5 – Identifying a protocol for file transfer

Question: A web developer needs to upload files to a web server. State which protocol would be used for this purpose. [1 mark]

Model answer:
FTP (File Transfer Protocol) **(1)**.

Exam tip: "Upload/download files" is the trigger phrase for FTP. This appeared in the 2024 exam. Do not confuse with HTTP which is for web pages.

Worked Example 6 – Comparing HTTP and HTTPS

Question: Explain why a bank's website uses HTTPS instead of HTTP. [2 marks]

Model answer:
HTTPS encrypts the data transmitted between the user and the website **(1)**. This protects sensitive information like passwords and account details from being intercepted and read **(1)**.

Exam tip: Always mention encryption and then explain why it matters — protecting sensitive data from interception.

Worked Example 7 – Email protocols

Question: State the protocol used to send an email. [1 mark]

Model answer:
SMTP **(1)**.

Exam tip: SMTP = Sending. POP3/IMAP = Receiving. This is a common exam question — know which protocol does which.

Worked Example 8 – Comparing POP3 and IMAP

Question: A user wants to access their email on multiple devices (phone, tablet, and laptop). Explain why IMAP would be more suitable than POP3 for this user. [2 marks]

Model answer:
IMAP keeps the email stored on the server **(1)**.
This means the email is synchronised across all devices, so changes (like marking as read) appear on all devices **(1)**.
Also acceptable for the second mark: POP3 downloads email to one device and deletes from server, so other devices would not receive it.

Exam tip: The key difference is where email is stored: POP3 downloads and deletes from server; IMAP keeps on server and syncs. Multiple devices = IMAP.

How email protocols work together

When you send and receive email, multiple protocols work together:

Action	Protocol used	What happens
You click "Send"	SMTP	Your email client sends the message to your email server
Server delivery	SMTP	Your server forwards the message to the recipient's server

Action	Protocol used	What happens
Recipient checks email	POP3 or IMAP	Recipient's email client retrieves the message from their server

Key point: Sending always uses SMTP. Receiving uses either POP3 or IMAP depending on how the email client is configured.

3. TCP/IP layers

The TCP/IP model organises network protocols into four layers. Each layer has a specific function and handles a different aspect of data transmission. Understanding what each layer does is essential for exam questions.

3.1 The four-layer model

The TCP/IP model divides network communication into four layers. Data passes down through the layers when sending, and up through the layers when receiving:

```
SENDING RECEIVING │ ▲ ▼ │ ┌──────────────────┐ ┌──────────────────┐ │ 4.
APPLICATION │ User data (web page) │ 4. APPLICATION │ │ HTTP, FTP... │ │
HTTP, FTP... │ └──────────┬───────┘ └──────────▲───────┘ │ │ ▼ │
┌──────────────────┐ ┌──────────────────┐ │ 3. TRANSPORT │ Splits into packets
│ 3. TRANSPORT │ │ TCP │ │ TCP │ └──────────┬───────┘ └──────────▲───────┘ │ │
▼ │ ┌──────────────────┐ ┌──────────────────┐ │ 2. INTERNET │ Adds IP
addresses │ 2. INTERNET │ │ IP │ │ IP │ └──────────┬───────┘
└──────────▲───────┘ │ │ ▼ │ ┌──────────────────┐ ┌──────────────────┐ │ 1.
LINK │ Converts to signals │ 1. LINK │ │ Ethernet/WiFi│ │ Ethernet/WiFi│
└──────────┬───────┘ └──────────▲───────┘ │ │ └──────────────── PHYSICAL MEDIUM
──────────────┘ (cables, radio waves)
```

3.2 Layer 4: Application layer

The application layer is the **top layer** that interfaces with the user and their applications.

Function

- Provides network services to applications (web browsers, email clients)
- Handles the specific format of requests and responses for each service
- Where the user's data originates

Protocols at this layer

HTTP, HTTPS, FTP, SMTP, POP3, IMAP

Exam phrase

"The application layer provides services to user applications and handles the format of data for specific services."

3.3 Layer 3: Transport layer

The transport layer manages **end-to-end communication** between devices.

Function

- Splits data into smaller packets for transmission
- Numbers packets so they can be reassembled in order
- Manages connections and ensures reliable delivery

- Numbers packets so they can be reassembled in order at the destination

Protocols at this layer

TCP (Transmission Control Protocol)

Exam phrase

"The transport layer splits data into packets, manages connections, and ensures packets are delivered reliably and in order."

3.4 Layer 2: Internet layer

The internet layer handles **addressing and routing** packets across networks.

Function

- Adds source and destination IP addresses to packets
- Routes packets across multiple networks to reach the destination
- Determines the best path for packets to travel

Protocols at this layer

IP (Internet Protocol)

Exam phrase

"The internet layer adds IP addresses to packets and routes them across networks to the destination."

3.5 Layer 1: Link layer

The link layer handles **physical transmission** on the local network.

Function

- Converts packets into signals (electrical, optical, or radio)
- Transmits data across the physical medium (cables or wireless)
- Uses MAC addresses to identify devices on the local network
- Handles access to the physical network medium

Protocols at this layer

Ethernet (wired), Wi-Fi (wireless)

Exam phrase

"The link layer converts packets into signals for transmission on the physical network and uses MAC addresses for local delivery."

Exam tip: A common question is to describe what happens at each layer. Remember: Application = user services, Transport = packets and reliability, Internet = IP addressing and routing, Link = physical signals and MAC addresses.

3.6 How the layers work together

When you request a web page, data passes through all four layers:

Step	Layer	What happens
1	Application	Browser creates an HTTP request for the web page

Step	Layer	What happens
2	Transport	TCP splits the request into packets, adds sequence numbers
3	Internet	IP adds source and destination IP addresses, determines route
4	Link	Data converted to electrical/radio signals and transmitted via Ethernet/Wi-Fi
5	*Packets travel across the internet through routers...*	
6	Link (server)	Signals received and converted back to packets
7	Internet (server)	IP addresses checked, packets delivered to correct device
8	Transport (server)	Packets reassembled in order, sent to correct application
9	Application (server)	Web server processes HTTP request and sends back the page

3.7 Benefits of the layered model

Organising protocols into layers has several advantages:

- **Modularity:** Each layer can be developed and updated independently
- **Standardisation:** Different manufacturers can create products that work at one layer without affecting others
- **Troubleshooting:** Problems can be isolated to a specific layer
- **Flexibility:** Protocols at one layer can be changed without affecting other layers (e.g., switching from Ethernet to Wi-Fi)

Example of layer flexibility

Consider switching from a wired to wireless connection:

- Only the **link layer** changes (Ethernet → Wi-Fi)
- The internet layer (IP) works exactly the same
- The transport layer (TCP) works exactly the same
- The application layer (HTTP) works exactly the same

This is why you can browse the same websites whether connected via cable or Wi-Fi — the upper layers do not need to know or care how the physical connection works.

3.8 Encapsulation

As data passes down through the layers, each layer adds its own header information. This process is called **encapsulation**:

Layer	Information added	Result
Application	Application-specific data (HTTP request)	Application data
Transport	Port numbers, sequence numbers	Segment
Internet	Source and destination IP addresses	Packet

Layer	Information added	Result
Link	MAC addresses, error checking	Frame

At the receiving end, each layer removes its header and passes the data up to the next layer — this is called **decapsulation**.

> **Note:** The terms segment, packet, and frame are sometimes used interchangeably in everyday usage, but technically they refer to data at different layers. For the exam, "packet" is acceptable for general use.

3.9 Complete example: Loading a web page

Let's trace what happens when you type "www.example.com" into your browser, showing how all layers work together:

```
YOUR COMPUTER WEB SERVER ──────────── ────────── 1. Application Layer
(HTTP) Browser creates HTTP request for the web page │ ▼ 2. Transport Layer
(TCP) Request split into packets Sequence numbers added │ ▼ 3. Internet
Layer (IP) Source and destination IP addresses added Route determined │ ▼ 4.
Link Layer (Ethernet/Wi-Fi) Converted to electrical/radio signals
Transmitted to router │ └──────────► INTERNET ──────────► (routed through │
multiple networks) ▼ 4. Link Layer Signals received │ ▼ 3. Internet Layer IP
addresses checked │ ▼ 2. Transport Layer Packets reassembled │ ▼ 1.
Application Layer HTTP request processed Web page sent back
```

The response (the web page) follows the same path in reverse. The whole process typically takes less than a second, even though data may travel through dozens of routers across multiple countries.

Worked Example 9 – Describing the link layer

Question: Describe the function of the link layer in the TCP/IP model. [2 marks]

Model answer:
The link layer converts data/packets into signals (electrical or radio) for transmission **(1)**.
It handles physical transmission on the local network using protocols like Ethernet or Wi-Fi **(1)**.
Also acceptable: Uses/adds MAC addresses for source/destination on local network.

> **Exam tip:** The 2024 mark scheme awarded marks for: converts to signals, transmission on media/cable/Wi-Fi, and using MAC addresses. Include two of these for 2 marks.

Worked Example 10 – Describing the transport layer

Question: Describe the function of the transport layer in the TCP/IP model. [2 marks]

Model answer:
The transport layer splits data into packets for transmission **(1)**.
It ensures reliable delivery by checking that packets arrive correctly and in order **(1)**.
Also acceptable: Manages connections between source and destination / Numbers packets for reassembly.

Exam tip: Transport layer key words: packets, reliable delivery, order, connections, sequence numbers. Pick two distinct points for 2 marks.

Worked Example 11 – Identifying layer protocols

Question: State which layer of the TCP/IP model each of the following protocols operates at:
(a) HTTP [1 mark]
(b) IP [1 mark]
(c) Ethernet [1 mark]

Model answer:
(a) Application layer **(1)**
(b) Internet layer **(1)**
(c) Link layer **(1)**

Exam tip: Know which protocols belong to which layer: Application (HTTP, HTTPS, FTP, SMTP, POP3, IMAP), Transport (TCP), Internet (IP), Link (Ethernet, Wi-Fi).

Worked Example 12 – Explaining how layers work together

Question: Describe how the transport and internet layers work together when sending data across a network. [4 marks]

Model answer:
Transport layer:
The transport layer receives data from the application layer and splits it into smaller packets **(1)**.
It numbers the packets so they can be reassembled in the correct order at the destination **(1)**.
Internet layer:

The internet layer adds the source and destination IP addresses to each packet **(1)**.
It determines the route and forwards packets across networks towards the destination **(1)**.

Exam tip: For questions about layers working together, describe what each layer does in sequence — transport prepares packets, internet addresses and routes them. Show you understand the flow of data through layers.

Worked examples summary

The table below summarises the 12 worked examples covered in this topic, organised by section and difficulty level.

#	Topic	Question type	Marks	Key skill
1	Definition	Define protocol	2	Set of rules + data transmission
2	Purpose	Explain why needed	2	Common rules for different devices
3	Standardisation	State benefit	1	Interoperability
4	Link protocols	Name wired/wireless	2	Ethernet = wired, Wi-Fi = wireless
5	FTP	Identify protocol	1	File transfer = FTP
6	HTTPS	Explain vs HTTP	2	Encryption protects data
7	SMTP	State sending protocol	1	SMTP = sending email
8	POP3 vs IMAP	Explain difference	2	Server storage and sync
9	Link layer	Describe function	2	Signals and physical transmission
10	Transport layer	Describe function	2	Packets and reliable delivery
11	Layer protocols	Identify layers	3	Protocol-to-layer mapping
12	Layers together	Describe interaction	4	Transport + Internet roles

Quick reference

Key terms and definitions

Term	Definition (mark-scheme style)
Protocol	A set of rules that govern how data is transmitted and received across a network
TCP/IP	The foundation protocol suite for internet communication; TCP handles reliable delivery, IP handles addressing and routing
HTTP	HyperText Transfer Protocol; used to transfer web pages (not encrypted)
HTTPS	HTTP Secure; transfers web pages using encryption to protect data
FTP	File Transfer Protocol; used to upload and download files between computers
SMTP	Simple Mail Transfer Protocol; used to send outgoing email
POP3	Post Office Protocol; downloads email to device and usually deletes from server
IMAP	Internet Message Access Protocol; keeps email on server and synchronises across devices
Ethernet	Protocol for wired LAN communication using cables
Wi-Fi	Protocol for wireless LAN communication using radio waves
IP address	A unique numerical address that identifies a device on a network
MAC address	A unique hardware address assigned to a network interface card for local network identification
Packet	A small unit of data transmitted across a network, containing payload and header information

Protocol purposes at a glance

Protocol	Purpose	Layer
HTTP	Transfer web pages (unsecured)	Application
HTTPS	Transfer web pages (encrypted/secure)	Application
FTP	Transfer files between computers	Application
SMTP	Send outgoing email	Application
POP3	Receive email (download to device)	Application
IMAP	Receive email (keep on server, sync)	Application
TCP	Reliable packet delivery and connections	Transport

Protocol	Purpose	Layer
IP	Addressing and routing packets	Internet
Ethernet	Wired LAN communication	Link
Wi-Fi	Wireless LAN communication	Link

TCP/IP layer functions

Layer	Number	Function	Protocols
Application	4	Provides services to user applications; handles data format for specific services	HTTP, HTTPS, FTP, SMTP, POP3, IMAP
Transport	3	Splits data into packets; ensures reliable, ordered delivery; manages connections	TCP
Internet	2	Adds IP addresses; routes packets across networks to destination	IP
Link	1	Converts packets to signals; handles physical transmission; uses MAC addresses	Ethernet, Wi-Fi

Email protocols comparison

Aspect	SMTP	POP3	IMAP
Direction	Sending	Receiving	Receiving
Server storage	N/A	Deleted after download	Kept on server
Multiple devices	N/A	No (one device)	Yes (synced)
Offline access	N/A	Yes (downloaded)	Limited

Typical pitfalls

Pitfall 1 — Confusing HTTP and FTP.
Wrong: "HTTP is used to transfer files to a server."
Right: HTTP is for **web pages**. FTP (File Transfer Protocol) is for **uploading/downloading files**.

Pitfall 2 — Forgetting the difference between HTTP and HTTPS.
Wrong: "HTTP and HTTPS are the same, just different versions."

Right: HTTPS uses **encryption** to protect data during transmission. HTTP does not encrypt data.

Pitfall 3 — Mixing up POP3 and IMAP.
Wrong: "POP3 keeps email on the server so you can access it from multiple devices."
Right: POP3 **downloads and deletes** from server (one device). IMAP **keeps on server** and syncs (multiple devices).

Pitfall 4 — Saying SMTP receives email.
Wrong: "SMTP is used to receive incoming email."
Right: SMTP is for **sending** email only. POP3 or IMAP is used to **receive** email.

Pitfall 5 — Confusing Ethernet and internet.
Wrong: "Ethernet is the protocol for the internet."
Right: Ethernet is for **wired local network (LAN)** communication. TCP/IP is the protocol suite for the internet.

Pitfall 6 — Wrong layer assignments.
Wrong: "HTTP operates at the transport layer."
Right: HTTP is an **application layer** protocol. TCP operates at the transport layer.

Pitfall 7 — Confusing IP addresses and MAC addresses.
Wrong: "MAC addresses are used to route packets across the internet."
Right: IP addresses route packets across networks. **MAC addresses** identify devices on the local network only.

Pitfall 8 — Incomplete protocol definition.
Wrong: "A protocol is how computers talk."
Right: "A protocol is a **set of rules** that defines how data is **transmitted and received** across a network." Include both elements.

Pitfall 9 — Missing the link layer's role.
Wrong: "The link layer just sends packets."
Right: The link layer **converts packets to signals** (electrical/radio), handles **physical transmission**, and uses **MAC addresses** for local delivery.

Pitfall 10 — Thinking Wi-Fi is internet access.
Wrong: "Wi-Fi gives you internet access."
Right: Wi-Fi is a **wireless LAN protocol** for local network communication. Internet access requires a router connected to an ISP — Wi-Fi just connects you to that

router wirelessly.

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